

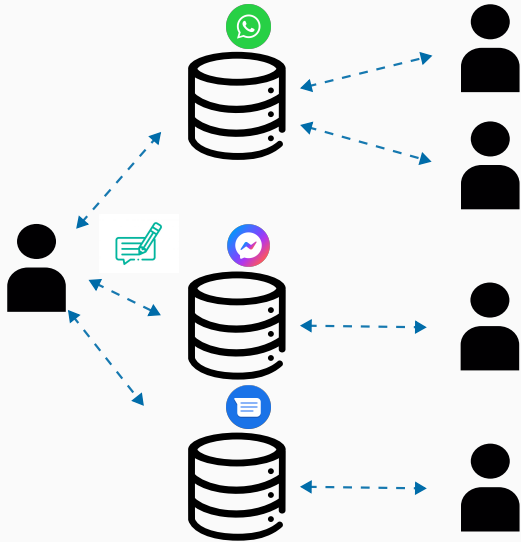
SolidSessionBench: Realistic Query Sequences for User-Oriented Decentralized Environments

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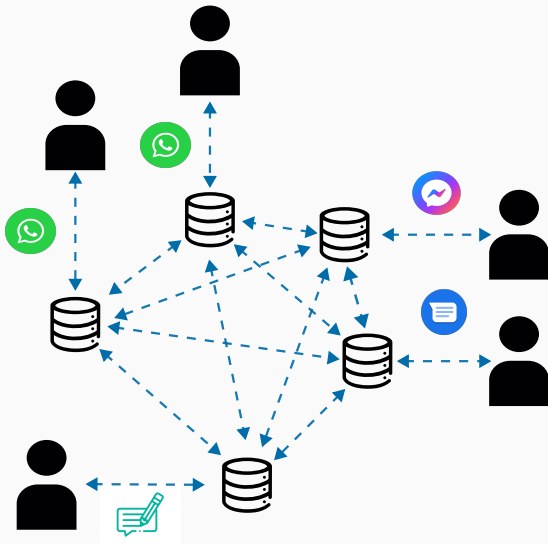
Department of Electronics and Information Systems, Ghent University – Imec

The Motivation for Decentralized Personal Data Storage



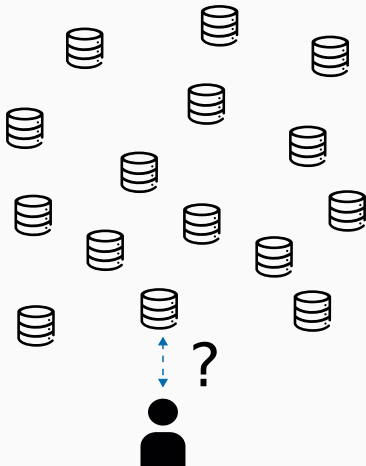
- Each application has its own data
- Stifles innovation
- Causes vendor lock-in

The Motivation for Decentralized Personal Data Storage



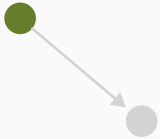
- Each application accesses a common decentralized data environment
- Easy to switch vendors
- Promotes innovation

Querying Structured Decentralized Environments



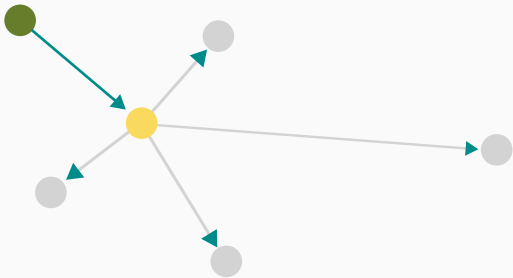
- How do we discover and query these sources?

Link Traversal: Seed Document



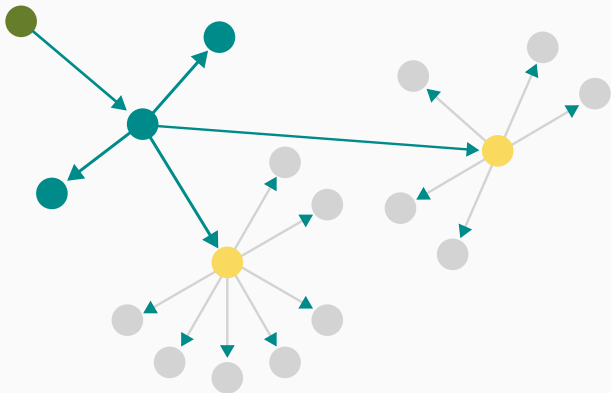
- Link Traversal starts from seed documents (URIs)
- These are provided by the user or specified in the query

Link Traversal: Traversal



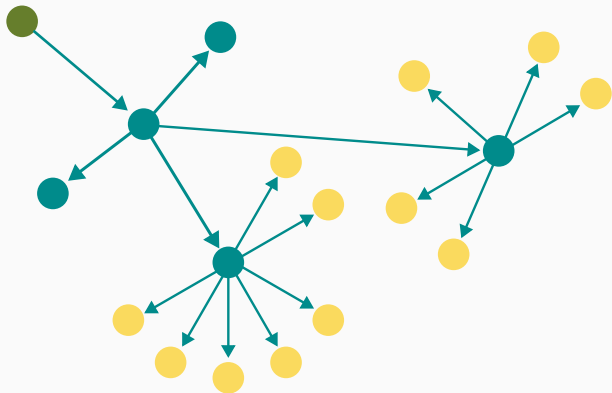
- New URIs are extracted from the seed document
- URIs are extracted in accordance with reachability criteria
- Uses the follow-your-nose principle

Link Traversal: Traversal



- New URIs are dereferenced and the process is repeated

Link Traversal: Termination

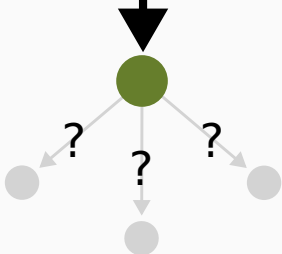


- This continues until all links are dereferenced
- Continuously produces results

Query Optimization for Link Traversal

```
SELECT * WHERE {  
  <seedUri> <ex:p1> ?o1.  
  <seedUri> <ex:p2> ?o2.  
  ...  
}
```

Extract
Seed document

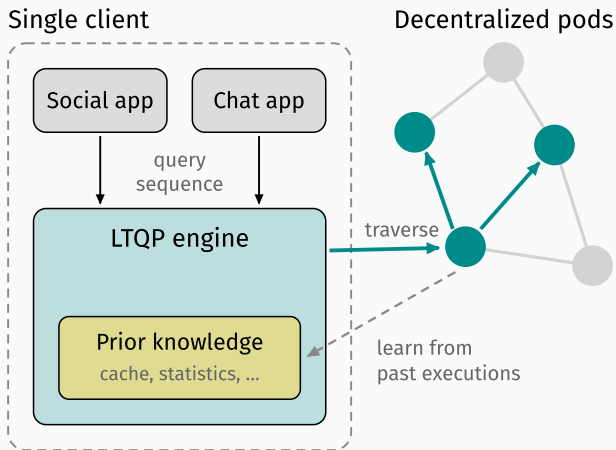


The query (partly) determines:

- The queried data
- The topology of the queried data
- The query-relevant documents

Result: Limited prior knowledge for query optimization

Query Optimization for Link Traversal: Client-side Engines

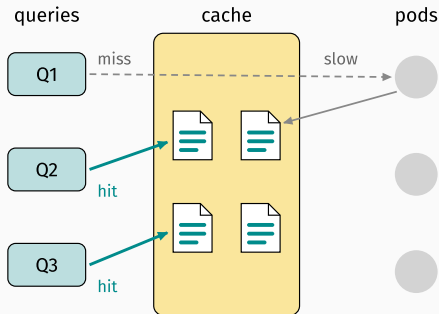


- Engine instances serve a single client
- Can we exploit client-specific usage patterns and 'personalize' optimization
- Gain prior knowledge by looking at past executions

Problem Statement

Evaluating the performance of client-specific optimization techniques in realistic, user-specific scenarios

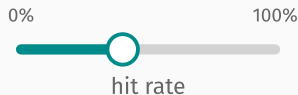
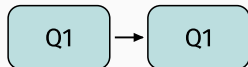
Evaluating Personalized Optimization through Caching



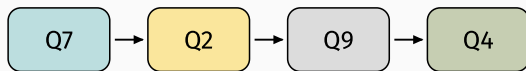
- Personalized optimization relies on data from preceding queries
 - Caching is a direct, well-studied instance of this (Hartig 2011)
- ⇒ How does the caching literature evaluate such techniques?

Existing Evaluation Approaches

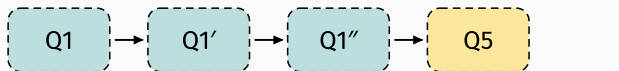
Micro benchmark



Random grouping



Real-world query log



- Micro benchmarks isolate single variables (Hartig 2011; Nicholson et al. 2024)
- End-to-end benchmarks group unrelated queries (Bizer and Schultz 2011; Martin et al. 2010)
- Real-world query logs are the gold standard (Saleem et al. 2015), but none exist for LTQP

Goal

- Create a unified benchmark for client-specific approaches
- Simulate real-world user behavior in reproducible manner

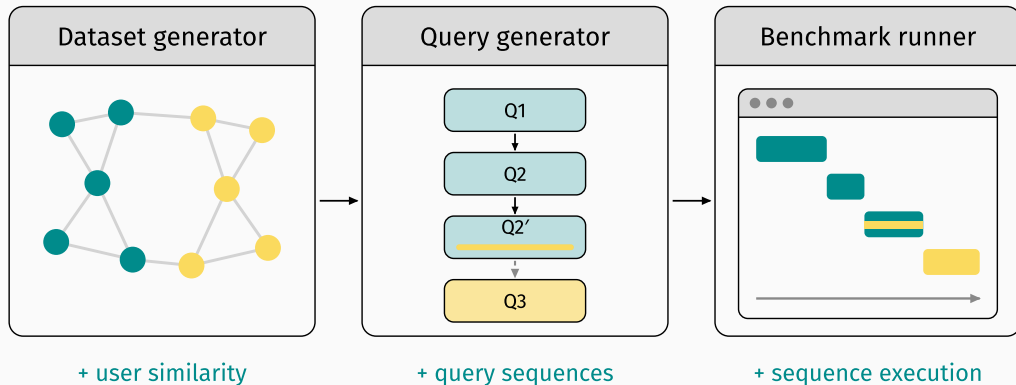
Simulated Solid Environment: SolidBench (Taelman and Verborgh 2023)

- Based on the realistic LDBC Social Network Benchmark
- Data is stored in vaults simulating social media application

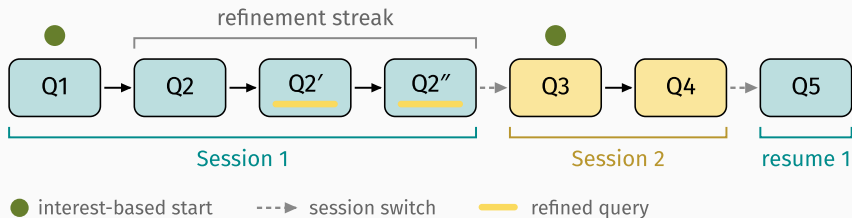
Data attributes

- Generates 158,233 RDF files
- Across 1,531 data vaults
- containing a total of 3,556,159 triples.

Simulation Methodology: Extending SolidBench



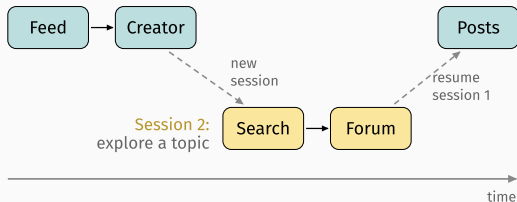
Simulation Methodology: Generating a Query Sequence



- Each sequence starts from a randomly selected base user
- Sessions chain dependent queries, refinement streaks modify them step-by-step
- A single seeded random generator keeps sequences reproducible

Usage Pattern: Logical Sessions

Session 1: follow a creator



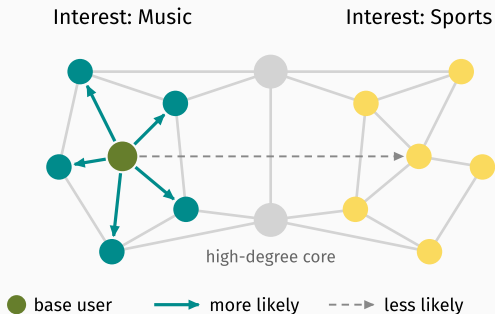
Observed (Meiss et al. 2009)

- Browsing groups into goal-oriented sessions
- New sessions in $\sim 15\%$ of clicks

Simulated

- Next query instantiated from previous results
- Switch to a new session or resume an old one

Usage Pattern: User Interests



Observed (Mislove et al. 2007; Mitzlaff et al. 2013)

- Users form communities around shared interests
- Shared interests drive more interactions

Simulated

- User similarity on interests and studies
- Similar users are more likely to start a session

Usage Pattern: Query Refinement

```
SELECT * WHERE {
```

```
  ?m :hasCreator <alice> .
```

```
  ?m :content ?text .
```

```
  FILTER(CONTAINS(?text, "jazz"))
```

```
  OPTIONAL { ?m :hasTag ?tag }
```

```
}
```

3 Shifting query intent: new seed

0 Base template

1 Result refinement: + FILTER

2 Result extension: + OPTIONAL

Observed (Bonifati et al. 2020; Arenas et al. 2025)

- Queries refined step-by-step
- Mostly FILTER, UNION, OPTIONAL

Simulated

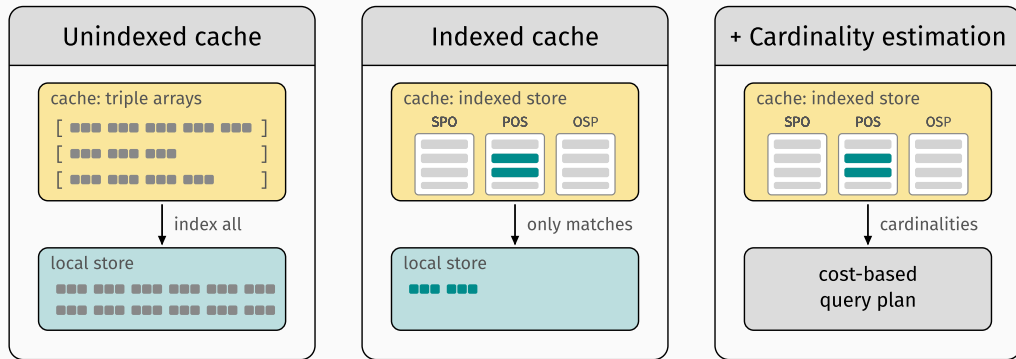
- Composable transformations on queries
- Randomly select sequence of transformations to add to query sequence

Validating the Benchmark: Client-side Caching

- Caching directly exploits data from past queries (Hartig 2011)
- Correlated query sequences should lead to cache hits
- Cache hit rates reveal whether our sequences behave as intended

⇒ Goal: validate the benchmark, not an exhaustive cache comparison

Caching Approaches



Implemented in Comunica (Taelman, Van Herwegen, et al. 2018), LRU eviction, three sizes: S (350k quads, ~10%), M (700k, ~20%), L (1.4M, ~40%)

Experimental Setup

Workload

- 8 generated sequences, 192 queries (discover and short templates)

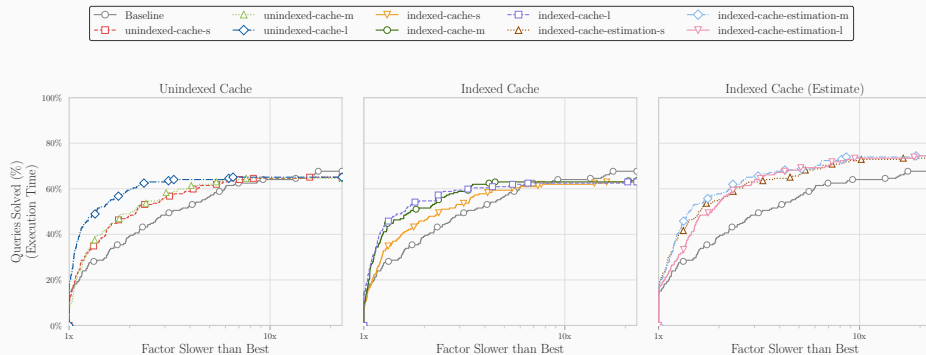
Execution

- Each sequence repeated 10 times, 70–90 ms injected network latency, 180 s timeout
- Cache scoped to a single sequence, no state shared between runs

Measured

- Cache hit rates, evictions, and speedup over no cache

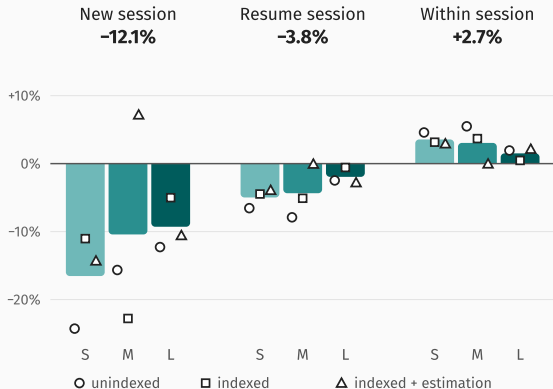
Results: All Caches Outperform No Caching



Performance profile (Dolan and Moré 2002): higher and further left is better

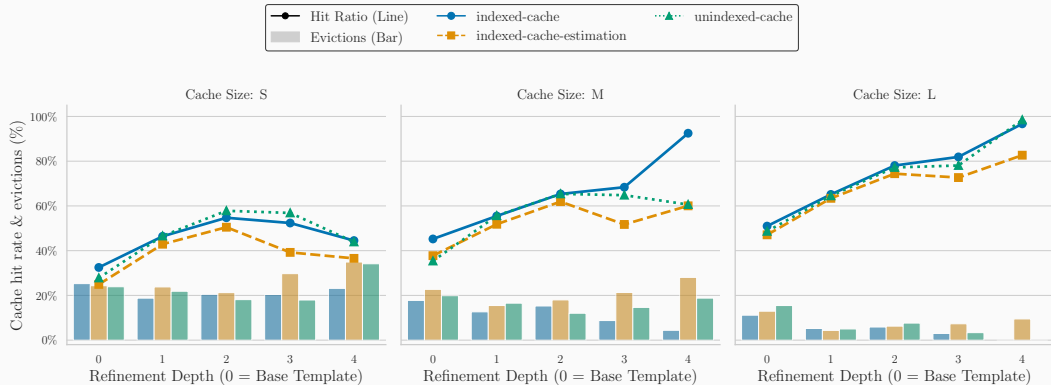
Cardinality estimation performs best and solves the most queries

Results: Session Switching Costs Cache Hits



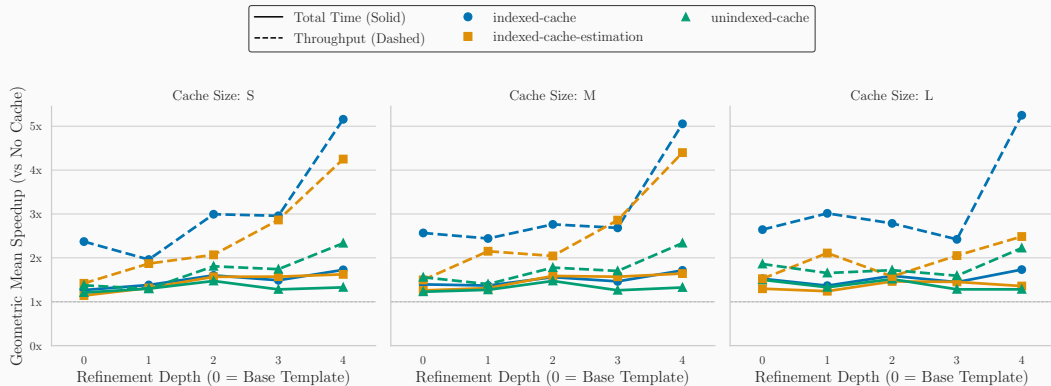
- Hit rate compared to the average of the same template
 - Strongest for small caches
- ⇒ Cache sizing and eviction depend on session behavior

Results: Refinement Reuses Prior Data



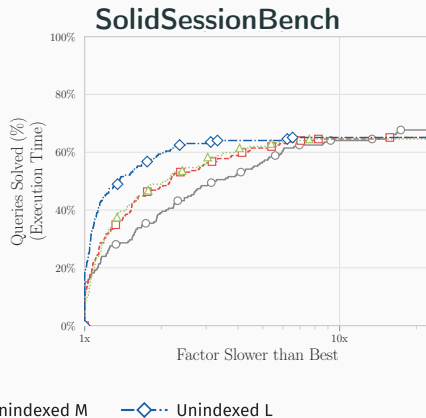
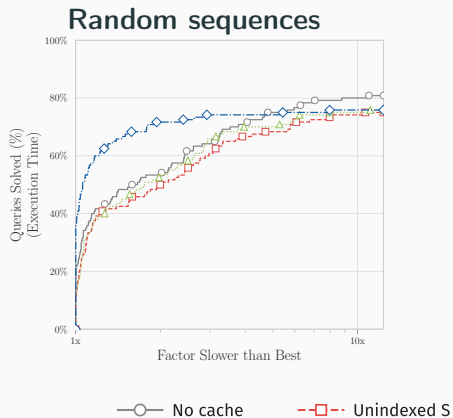
Hit rates rise with each refinement step, until small caches start thrashing

Results: Refinement Translates into Speedups



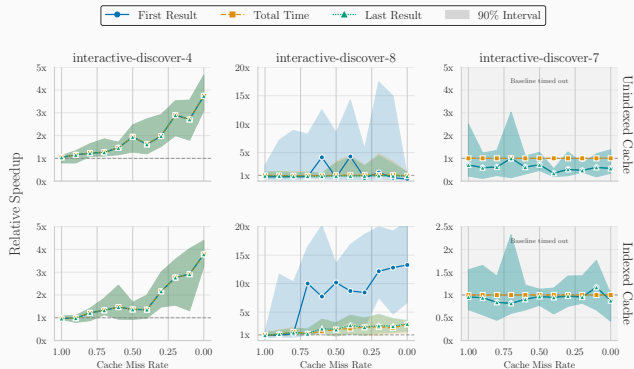
Up to $\sim 5\times$ throughput at refinement depth 4 with the indexed cache

Comparison: Random Sequences Mislead



Random: only large caches help vs. SolidSessionBench: small caches help too

Comparison: Micro Benchmarks Give Mixed Signals



- Each template run twice with simulated cache miss rates
 - 8 of 12 templates speed up with hit rate
 - Long-running queries: no speedup, or even slowdowns
- ⇒ Isolated queries do not show behavior in realistic sequences

Key Takeaways

- Realistic sequences reveal how sessions and refinement drive client-side performance
 - Random sequences and micro benchmarks miss these effects, or lead to the wrong conclusions
- ⇒ SolidSessionBench: a reproducible basis for evaluating client-side optimizations

-  Arenas, M. et al. (2025). **“Exploring Exploratory Querying”**. In: *Proceedings of the VLDB Endowment* 18.13, pp. 5731–5739.
-  Bizer, C. and A. Schultz (2011). **“The berlin sparql benchmark”**. In: *Semantic Services, Interoperability and Web Applications: Emerging Concepts*. IGI Global Scientific Publishing, pp. 81–103.
-  Bonifati, A., W. Martens, and T. Timm (2020). **“An analytical study of large SPARQL query logs”**. In: *The VLDB Journal* 29.2, pp. 655–679.
-  Dolan, E. D. and J. J. Moré (2002). **“Benchmarking optimization software with performance profiles”**. In: *Mathematical programming* 91.2, pp. 201–213.
-  Hartig, O. (2011). **“How Caching Improves Efficiency and Result Completeness for Querying Linked Data”**. In: *LDOW*.

-  Martin, M., J. Unbehauen, and S. Auer (2010). **“Improving the performance of semantic web applications with SPARQL query caching”**. In: *Extended Semantic Web Conference*. Springer, pp. 304–318.
-  Meiss, M. et al. (2009). **“What’s in a session: tracking individual behavior on the web”**. In: *Proceedings of the 20th ACM conference on Hypertext and hypermedia*. Torino Italy: ACM, pp. 173–182. DOI: 10.1145/1557914.1557946.
-  Mislove, A. et al. (2007). **“Measurement and analysis of online social networks”**. In: *Proceedings of the 7th ACM SIGCOMM conference on Internet measurement*, pp. 29–42.
-  Mitzlaff, F. et al. (2013). **“User-relatedness and community structure in social interaction networks”**. In: *arXiv preprint arXiv:1309.3888*.

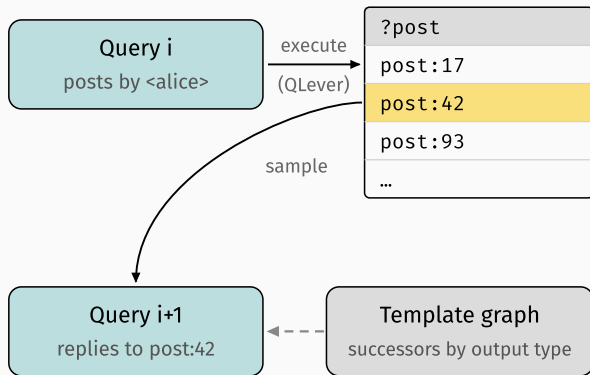
-  Nicholson, H., P. Chrysogelos, and A. Ailamaki (2024). “**HPCache: memory-efficient OLAP through proportional caching revisited**”. In: *The VLDB Journal* 33.6, pp. 1775–1791.
-  Saleem, M. et al. (2015). “**LSQ: the linked SPARQL queries dataset**”. In: *International semantic web conference*. Springer, pp. 261–269.
-  Taelman, R., J. Van Herwegen, et al. (2018). “**Comunica: a modular SPARQL query engine for the web**”. In: *International Semantic Web Conference*. Springer, pp. 239–255.
-  Taelman, R. and R. Verborgh (2023). “**Link traversal query processing over decentralized environments with structural assumptions**”. In: *International Semantic Web Conference*. Springer, pp. 3–22.

Simulation Methodology: Hyperparameters

Category	Parameter	Value
Log-normal (μ, σ)	Sequence length	3.0, 0.2
	Session length	1.7, 0.5
	Transition probability	−2.0, 0.25
Refinements	patternProbability	0.1
	min/maxRefinementLength	2, [3–5]
Instantiation	temperature	0.1
	maxLogits	5–10

A single seeded PRNG guarantees reproducible sequences

Simulation Methodology: Logical Query Sessions

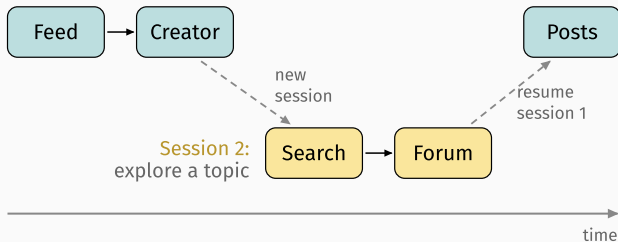


No results → fall back to interest-based instantiation

- Directed template graph:
output types map to valid next templates
- Instantiation values sampled from previous results
- Simulates “click-through” behavior

Simulation Methodology: Session Switching

Session 1: follow a creator



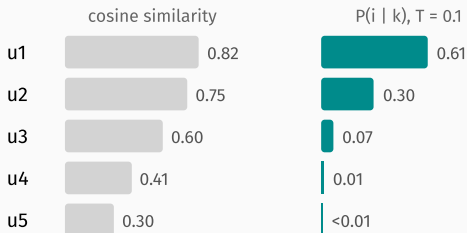
- Switch probability per sequence: log-normal, mean 15% (Meiss et al. 2009)
- Switch: new session or resume a previous one (50/50)
- No backtracking, balanced template distribution

Simulation Methodology: User Interest Modeling

Base user k



Top-M similar users



- Binary vector of interests and studies per user
- Cosine similarity to base user k , keep top- M
- Softmax selects the first query of each session:

$$P(i | k) = \frac{e^{s_{k,i}/T}}{\sum_{u \in C_{k,M}} e^{s_{k,u}/T}}$$

Simulation Methodology: Refinement Patterns

Behavior	Transformation	Choke points
Result refinement	$\pm$ FILTER, $\pm$ selective triple patterns	P.1, P.3
Result generalization	+ (non-)selective triple patterns	P.3, P.4
Result extension	+ triple patterns, + OPTIONAL/UNION	P.2–P.4
Query refinement	$\pm$ OPTIONAL/UNION	P.2
Shifting query intent	New instantiation values (seeds)	T.1, T.2

P.1 Filters

P.2 Filter push-down through OPTIONAL/UNION

P.3 Selective patterns

P.4 Non-selective patterns

Simulation Methodology: Refinements Impact Traversal

- **T.1** Predicate additions expand or contract the reachable sub-web
- **T.2** New starting points change the reachable documents
- **T.3** Changes the number of hops to relevant data (SolidBench L.1–L.6)
- **T.4** Alters usability of structural indexes, such as type indexes (L.8)

⇒ Custom refinement patterns can be defined in JSON